SecureLink

A self-hosted windows WireGuard VPN

Daniyar Abeuov CST-452 Grand Canyon University

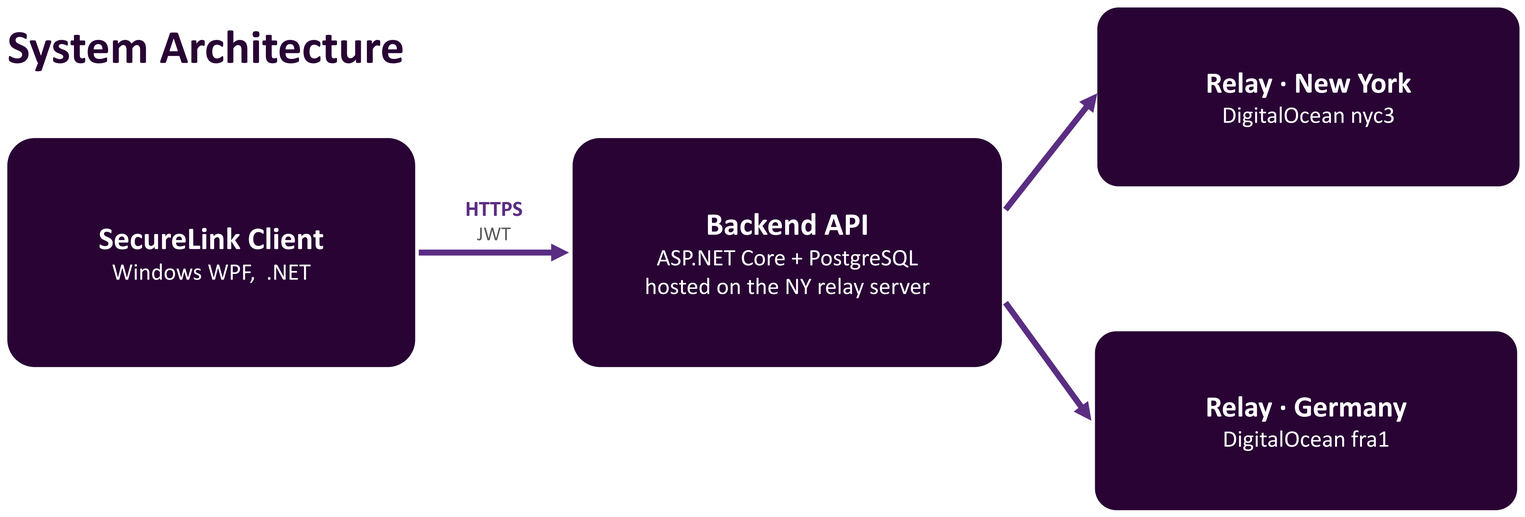
Private VPN, my own encrypted WireGuard relays in New York and Frankfurt, a Windows client that sets everything up in one click.

1 Project Description

SecureLink is a complete VPN application: desktop client, backend API and relay servers that I designed, built and deployed.

- A Windows (WPF) client app sets up an encrypted WireGuard tunnel to a relay in New York or Germany.
- An ASP.NET Core backend manages accounts, enforces one session per user.
- A kill switch blocks all traffic outside the tunnel, and the app cleans up automatically after a crash.

System Architecture



2 Project Justification

I personally don’t trust commercial VPNs with my traffic. Running your own VPN server saves money and ensures your data is not logged by a third-party provider.

- Control: my own servers, no third-party logging.
- Automation: key exchange, device setup and IP assignment happen for the user.
- Safety: the kill switch and crash recovery prevent silent leaks.
- Scale: built for a small group of users, hosted on low-cost servers.

3 Functional Requirements, Specifications & Standards

Functional requirements

- Register / log in (BCrypt, JWT)
- One active session per account
- Choose relay: New York or Germany
- Unique tunnel IP for every session
- Backend adds / removes devices on the relay
- Keys generated on the client only
- Connect, disconnect, live usage stats
- Kill switch blocks non-encrypted traffic outside the tunnel
- Cleanup after a crash or connection loss
- Relays built by Terraform + cloud-init

Product specifications

Client: Windows 10/11, .NET 8 WPF, WireGuard for Windows

Backend: ASP.NET Core 8, EF Core, PostgreSQL 14, Ubuntu

Relays: DigitalOcean nyc3 + fra1

Standards

- WireGuard
- TLS with SHA-256
- JWT; BCrypt hashing

4 Design Process

Plan Design documentation, user stories and functional requirements

Design Architecture, data dictionary, API

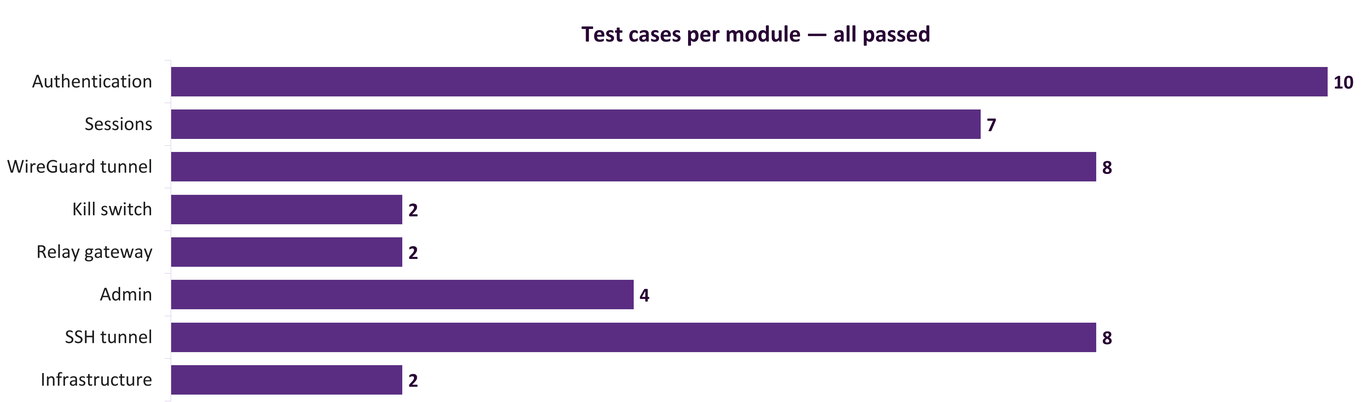
Build Agile sprints: 25 user stories with a traceability matrix

Test 43 test cases

Release Deployed and ready-to-use application

5 Data Summaries

43 / 43	100%	26	0.6 s
test cases passed	user stories done (25 / 25)	Defects/bugs found & fixed	fail-fast when a relay is down



Source code: github.com/AbeuovDaniyar/SecureLinkVPN